



NORTH SYDNEY BOYS HIGH SCHOOL

MATHEMATICS (EXTENSION 1)

2014 HSC Course Assessment Task 3 (Trial Examination)

Tuesday June 24, 2014

General instructions

- Working time – 2 hours.
(plus 5 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets + answer sheet used in the correct order within this paper and hand to examination supervisors.

SECTION I

- Mark your answers on the answer sheet provided (numbered as page 13)

SECTION II

- Commence each new question on a new page. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

STUDENT NUMBER: **# BOOKLETS USED:**

Class (please ✓)

☐ 12M3A – Mr Zuber

☐ 12M4A – Ms Ziaziaris

☐ 12M3B – Mr Berry

☐ 12M4B – Mr Lam

☐ 12M3C – Mr Lowe

☐ 12M4C – Mr Ireland

Marker's use only.

QUESTION	1-10	11	12	13	14	Total	%
MARKS	$\overline{10}$	$\overline{15}$	$\overline{15}$	$\overline{15}$	$\overline{15}$	$\overline{70}$	

Section I

10 marks

Attempt Question 1 to 10

Allow approximately 15 minutes for this section

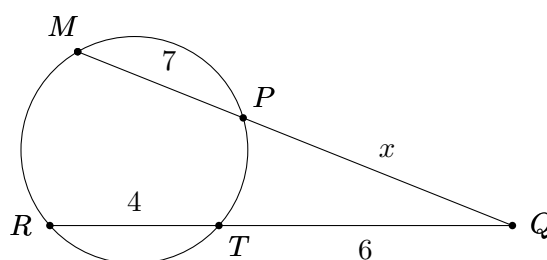
Mark your answers on the answer sheet provided.

Questions

Marks

1. What is the value of x if $MP = 7$, $RT = 4$, $TQ = 6$?

1



(A) $\frac{21}{2}$

(B) 5

(C) $\frac{24}{7}$

(D) 9

2. What is the remainder when $8x^3 + 4x^2 - 3x - 8$ is divided by $x - 1$?

1

(A) -1

(B) 1

(C) -9

(D) -8

3. Which of the following calculations is required to calculate the acute angle between the lines $x + y = 1$ and $2x - y = 1$?

1

(A) $\tan \theta = \left| \frac{-1 + 2}{1 + 2} \right|$

(B) $\tan \theta = \left| \frac{-1 + 2}{2} \right|$

(C) $\tan \theta = \left| \frac{-1 - 2}{1 - 2} \right|$

(D) $\tan \theta = \left| \frac{-1 - 2}{-2} \right|$

4. What is the general solution to $2 \sin \frac{x}{2} = -1$? 1

(A) $2n\pi - (-1)^n \frac{\pi}{3}$

(B) $n\pi - (-1)^n \frac{\pi}{6}$

(C) $n\pi - (-1)^n \frac{\pi}{3}$

(D) $2n\pi - (-1)^n \frac{\pi}{6}$

5. Solve: $x^2(x^2 - 9) - 10 > 0$. 1

(A) $-\sqrt{10} < x < \sqrt{10}$

(B) $x > \sqrt{10}, -1 < x < 1, x < -\sqrt{10}$

(C) $1 < x < \sqrt{10}, -\sqrt{10} < x < -1$

(D) $x > \sqrt{10}, x < -\sqrt{10}$

6. Which of the following integrals is equal to $\int \sin^2 4t \, dt$? 1

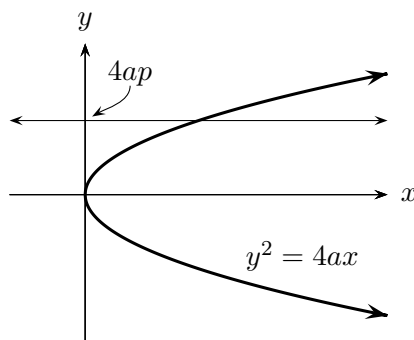
(A) $\int \frac{1 + \cos 8t}{2} \, dt$

(B) $\int \frac{1 - \cos 8t}{2} \, dt$

(C) $\int \frac{1 + \cos 4t}{2} \, dt$

(D) $\int \frac{1 - \cos 4t}{2} \, dt$

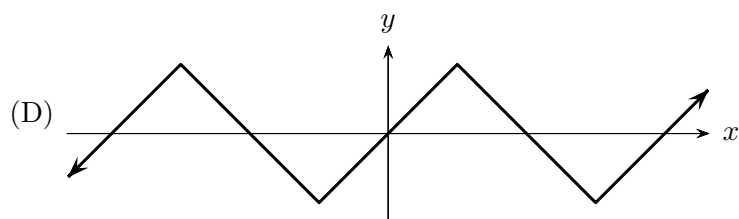
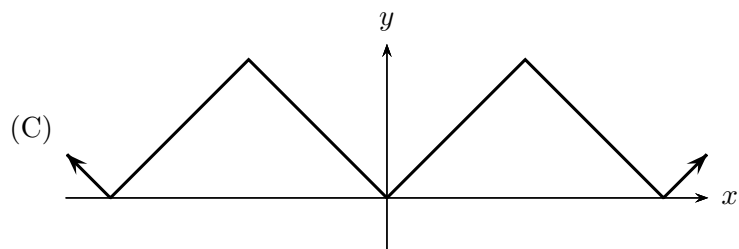
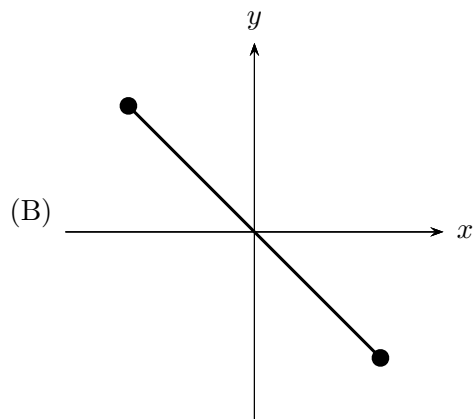
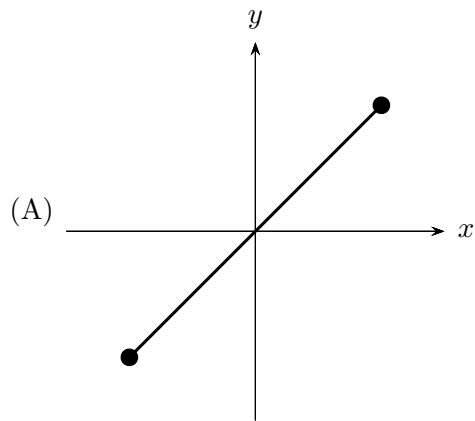
7. Which of the following integrals gives the volume of the solid formed when the area bound by $y^2 = 4ax$, the y axis and the line $y = 4ap$ is rotated about the y axis? 1



- (A) $\pi \int_0^{4ap} 4ax \, dx$
- (B) $\pi \int_0^{4ap} \frac{y^4}{16a^2} \, dx$
- (C) $\pi \int_0^{4ap} \frac{y^2}{16a} \, dy$
- (D) $\pi \int_0^{4ap} \sqrt{4ax} \, dx$
8. Evaluate: $\lim_{x \rightarrow 0} \frac{\sin^2 x}{2x^2}$. 1
- (A) $\frac{1}{2}$
- (B) 1
- (C) 2
- (D) ∞
9. A box contains 4 coins, one of which has two heads (no tails). If a coin is selected at random and then tossed, what is the probability that heads will show? 1
- (A) $\frac{3}{8}$ (B) $\frac{3}{4}$ (C) $\frac{5}{8}$ (D) $\frac{4}{5}$

10. Which of the following graphs represents $y = \cos(\cos^{-1} x)$?

1



Examination continues overleaf...

Section II

60 marks

Attempt Questions 11 to 14

Allow approximately 1 hour and 45 minutes for this section.

Write your answers in the writing booklets supplied. Additional writing booklets are available. Your responses should include relevant mathematical reasoning and/or calculations.

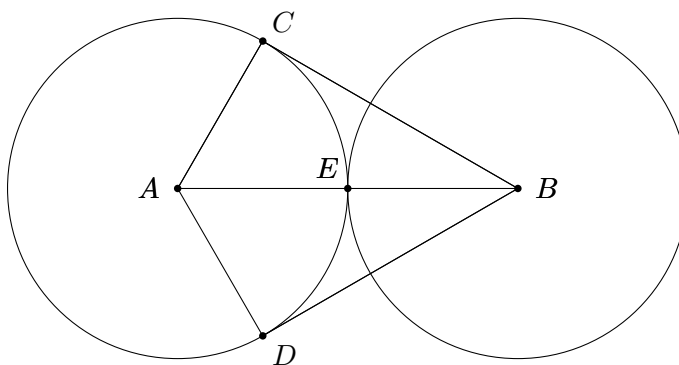
Question 11 (15 Marks)	Commence a NEW page.	Marks
(a) Differentiate $\cos^{-1} \frac{x}{3}$.		1
(b) Evaluate the following integrals:		1
i. $\int \frac{1}{\cos^2 x} dx$.		
ii. $\int_0^{\frac{1}{2}} \frac{2 dx}{1 + 4x^2}$.		3
(c) Evaluate $\int x\sqrt{1-x} dx$ by using the substitution $u = 1 - x$.		3
(d) If α , β and γ are roots of the equation $x^3 - 3x^2 - 5x - 7 = 0$, evaluate:		
i. $\alpha + \beta + \gamma$.		1
ii. $\alpha\beta\gamma$.		1
iii. $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$.		2
(e) If $\cos x \cos y = \frac{1}{4}(\sqrt{3} - \sqrt{2})$ and $\sin x \sin y = \frac{1}{4}(\sqrt{3} + \sqrt{2})$, find the smallest positive value of $(x + y)$ in radians.		3

Question 12 (15 Marks)

Commence a NEW page.

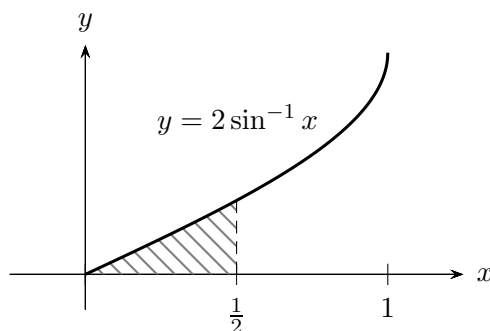
Marks

- (a) Prove that $\tan^{-1} \frac{1}{4} + \tan^{-1} \frac{3}{5} = \frac{\pi}{4}$. **3**
- (b) i. Express $\sqrt{2} \cos x - \sqrt{2} \sin x$ in the form $R \cos(x + \alpha)$, where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$. **2**
- ii. Hence sketch $y = \sqrt{2} \cos x - \sqrt{2} \sin x$ for $0 \leq x \leq 2\pi$. **2**
- iii. Hence find the value of k for which $\sqrt{2} \cos x - \sqrt{2} \sin x = k$ will have 3 solutions in the domain $0 \leq x \leq 2\pi$. **1**
- (c) Two circles of *equal radii* and centres A and B respectively, touch externally at E . BC and BD are tangents from B to the circle with centre A .



Copy or trace the diagram into your working booklet.

- i. Show that $BCAD$ is a cyclic quadrilateral. **2**
- ii. Show that E is the centre of the circle which passes through B , C , A and D . **1**
- iii. Show that $\angle CBA = \angle DBA = 30^\circ$. **1**
- (d) The diagram below shows $f(x) = 2 \sin^{-1} x$. The area enclosed between the curve, the line $x = \frac{1}{2}$ and the x axis is shaded.



- i. Find $f\left(\frac{1}{2}\right)$. **1**
- ii. Show that the area of the shaded region is given by **2**

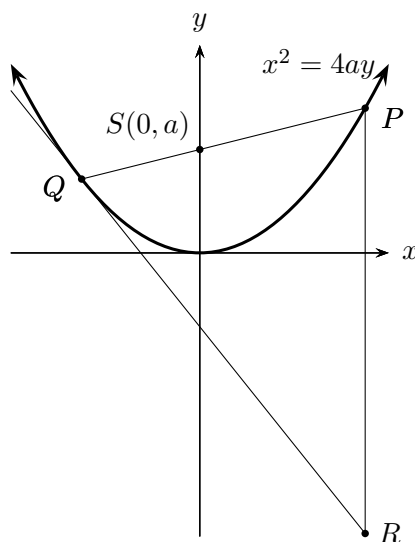
$$\frac{\pi}{6} + \sqrt{2} - 2 \text{ units}^2$$

Question 13 (15 Marks)

Commence a NEW page.

Marks

- (a) PQ is a focal chord of the parabola $x^2 = 4ay$, where $P(2ap, ap^2)$ and $Q(2aq, aq^2)$.
 QR is the tangent at Q and RP is parallel to the axis of the parabola.



- i. Show that $pq = -1$. 2
 - ii. Find the equation of the locus of R . 3
- (b) The acceleration of a particle is given by $\ddot{x} = 4x - 4$. Initially, $x = 6$ and $|v| = 8$.
- i. Show that $v^2 = 4x^2 - 8x - 32$. 2
 - ii. Find the set of possible values of x . 1
- (c) An object is removed from a freezer at -5°C and is placed in a room where the temperature is kept constant at 15°C . The temperature of the object T then changes according to the rule

$$\frac{dT}{dt} = k(15 - T)$$

where t is in minutes and k is a constant.

- i. Verify that $T = 15 - Ae^{-kt}$ satisfies this condition. 1
 - ii. Find the value of A . 1
 - iii. If initially the temperature of the object was increasing at 5°C per minute, find the value of k . 1
 - iv. Find, correct to the nearest second, the time it takes for the object's temperature to rise to 0°C . 1
- (d) Prove by induction that $9^{n+2} - 4^n$ is divisible by 5 for all positive integers n . 3

Question 14 (15 Marks)

Commence a NEW page.

Marks

- (a) The position of a particle moving along the x axis is given by $x = t^2 e^{2-t}$, where x is in metres and t is in seconds respectively.
- i. Find an expression for the velocity in terms of time. **1**
 - ii. Show that the particle is initially at rest. **1**
 - iii. Find the time for which the particle is next at rest. **1**
 - iv. What happens as time increases indefinitely? **1**
- (b) It is known that a solution exists near $x = 2$ for the equation $\log_e x = (x - 1)^2$. **2**
- Use one application of Newton's Method to find a better approximation to $x = 2$, correct to two decimal places.
- (c) Consider the function $y = \log_e \frac{2x}{2+x}$.
- i. Show that the domain of the function is **2**
- $$D = \{x : x < -2, x > 0\}$$
- ii. Find the value of x for which $y = 0$. **1**
 - iii. Show that $\frac{dy}{dx} = \frac{2}{x(2+x)}$ and hence show that the function is increasing **2**
for all x in the domain.
 - iv. Are there any points of inflexion? Justify your answer. **2**
 - v. Sketch the graph of the function. **2**

End of paper.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1} + C, \quad n \neq -1; \quad x \neq 0 \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x + C, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax} + C, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax + C, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax + C, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax + C, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax + C, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + C, \quad a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right) + C, \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right) + C$$

NOTE: $\ln x = \log_e x$, $x > 0$

Answer sheet for Section I

Mark answers to Section I by fully blackening the correct circle, e.g. “●”

STUDENT NUMBER:

Class (please ✓)

☐ 12M3A – Mr Zuber

☐ 12M4A – Ms Ziariaris

☐ 12M3B – Mr Berry

☐ 12M4B – Mr Lam

☐ 12M3C – Mr Lowe

☐ 12M4C – Mr Ireland

- | | | | | |
|-------------|-------------------------|-------------------------|-------------------------|-------------------------|
| 1 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 2 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 3 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 4 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 5 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 6 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 7 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 8 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 9 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 10 – | ☐ A | ☐ B | ☐ C | ☐ D |

① B ② B ③ C ④ A ⑤ D ⑥ B ⑦ B ⑧ A ⑨ C ⑩ A

$$\begin{aligned}
 1. \quad (x+7)x &= 10 \times 6 \\
 x^2 + 7x - 60 &= 0 \\
 (x+12)(x-5) &= 0 \\
 x &= 5
 \end{aligned}$$

$$\begin{aligned}
 2. \quad 8(1) + 4(1) - 3(1) - 8 \\
 &= 1
 \end{aligned}$$

$$\begin{aligned}
 3. \quad \tan \theta &= \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \\
 &= \left| \frac{-1 - 2}{1 + (-1)(2)} \right| \\
 &= \left| \frac{-1 - 2}{1 - 2} \right|
 \end{aligned}$$

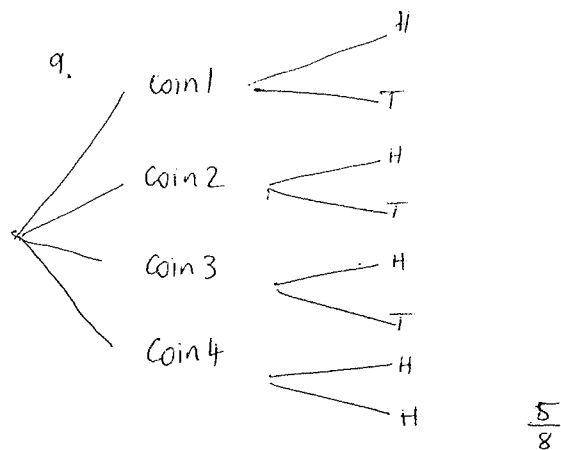
$$\begin{aligned}
 4. \quad \sin \frac{x}{2} &= -\frac{1}{2} \\
 \frac{x}{2} &= n\pi + (-1)^n \left(-\frac{\pi}{6} \right) \\
 x &= 2n\pi - (-1)^n \left(\frac{\pi}{3} \right)
 \end{aligned}$$

$$\begin{aligned}
 5. \quad x^2(x^2 - 9) - 10 &> 0 \\
 x^4 - 9x^2 - 10 &> 0 \\
 (x^2 - 10)(x^2 + 1) &> 0 \\
 (x - \sqrt{10})(x + \sqrt{10})(x^2 + 1) &> 0 \\
 x > \sqrt{10}, x < -\sqrt{10}
 \end{aligned}$$

$$6. \int \frac{1 - \cos 8t}{2} dt$$

$$\begin{aligned}
 7. \quad \pi \int_0^{4ap} x^2 dy \\
 \pi \int_0^{4ap} \left(\frac{y^2}{4a} \right)^2 dy
 \end{aligned}$$

$$\begin{aligned}
 8. \quad \lim_{x \rightarrow 0} \frac{\sin x \sin x}{2x \cdot x} \\
 &= \frac{1}{2}
 \end{aligned}$$



$$\begin{aligned}
 10. \quad \text{Plot points} \\
 \cos(\cos^{-1} 1) &= 1 \quad \text{Restricted because} \\
 \cos(\cos^{-1} -1) &= -1 \quad \cos^{-1} \text{ first.}
 \end{aligned}$$

11. (a) Let $y = \cos^{-1}\left(\frac{x}{3}\right)$

$$\frac{dy}{dx} = \frac{-1}{\sqrt{9-x^2}}$$

(b) (i) $\int \frac{1}{\cos^2 x} dx$

$$= \int \sec^2 x dx$$

$$= \tan x + C$$

(ii) $\int_0^{\frac{1}{2}} \frac{2dx}{1+4x^2}$

$$= 2 \int_0^{\frac{1}{2}} \frac{dx}{1+4x^2}$$

$$= \frac{2}{4} \int_0^{\frac{1}{2}} \frac{dx}{\frac{1}{4} + x^2}$$

$$= \frac{1}{2} \int_0^{\frac{1}{2}} \frac{dx}{\left(\frac{1}{2}\right)^2 + x^2}$$

$$= \frac{1}{2} \times 2 \left[\tan^{-1} 2x \right]_0^{\frac{1}{2}}$$

$$= \tan^{-1} 1 - \tan^{-1} 0$$

$$= \frac{\pi}{4}$$

OR $2 \int_0^{\frac{1}{2}} \frac{dx}{1+4x^2}$

$$= 2 \times \frac{1}{2} \left[\tan^{-1} 2x \right]_0^{\frac{1}{2}}$$

(c) $\int x \sqrt{1-x} dx$

$$u = 1-x$$

$$= \int (1-u) u^{\frac{1}{2}} du \quad \begin{array}{l} \frac{du}{dx} = -1 \\ du = -dx \end{array}$$

$$= - \int (u^{\frac{1}{2}} - u^{\frac{3}{2}}) du$$

$$= \int (u^{\frac{3}{2}} - u^{\frac{1}{2}}) du$$

$$= \frac{2}{5} u^{\frac{5}{2}} - \frac{2u^{\frac{3}{2}}}{3} + C$$

$$= \frac{2}{5} (1-x)^{\frac{5}{2}} - \frac{2}{3} (1-x)^{\frac{3}{2}} + C$$

$$(d) \quad x^3 - 3x^2 - 5x - 7 = 0$$

$$(i) \quad \alpha + \beta + \gamma = 3$$

$$(ii) \quad \alpha\beta\gamma = 7$$

$$(iii) \quad \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma}$$
$$= \frac{-5}{7}$$

$$(e) \quad \cos(x+y) = \cos x \cos y - \sin x \sin y$$
$$= \frac{1}{4}(\sqrt{3}-\sqrt{2}) - \frac{1}{4}(\sqrt{3}+\sqrt{2})$$

$$\cos(x+y) = -\frac{1}{2}\sqrt{2}$$

$$\cos(x+y) = -\frac{1}{\sqrt{2}}$$

$$\therefore x+y = \pi - \frac{\pi}{4}$$

$$x+y = \frac{3\pi}{4}$$

12, a) Prove $\tan^{-1} \frac{1}{4} + \tan^{-1} \frac{3}{5} = \frac{\pi}{4}$

Let $\tan^{-1} \frac{1}{4} = \alpha$ and $\tan^{-1} \frac{3}{5} = \beta$

$\therefore \tan \alpha = \frac{1}{4}$ $\tan \beta = \frac{3}{5}$

$\therefore$ Prove $\alpha + \beta = \pi/4$

Now, $\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$

$$= \frac{\frac{1}{4} + \frac{3}{5}}{1 - \frac{1}{4} \cdot \frac{3}{5}}$$

$$= \frac{\frac{17}{20}}{\frac{17}{20}}$$

$\tan(\alpha + \beta) = 1$

$\therefore \alpha + \beta = \tan^{-1} 1$
 $= \pi/4$

$\therefore \tan^{-1} \frac{1}{4} + \tan^{-1} \frac{3}{5} = \frac{\pi}{4}$

(b) (i) $\sqrt{2} \cos x - \sqrt{2} \sin x = R \cos(x + \alpha)$

$R = \sqrt{(\sqrt{2})^2 + (\sqrt{2})^2}$

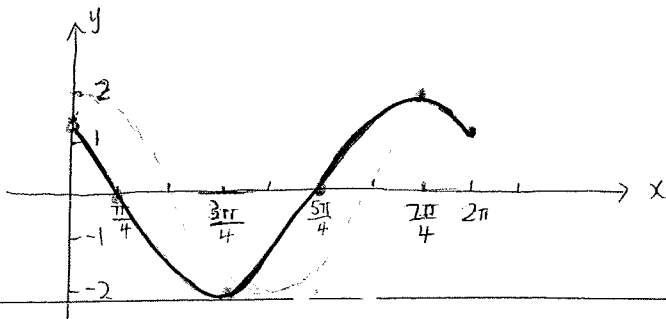
$R = 2$

$\alpha = \tan^{-1} \frac{\sqrt{2}}{\sqrt{2}}$

$\alpha = \tan^{-1} 1$

$\alpha = \frac{\pi}{4}$

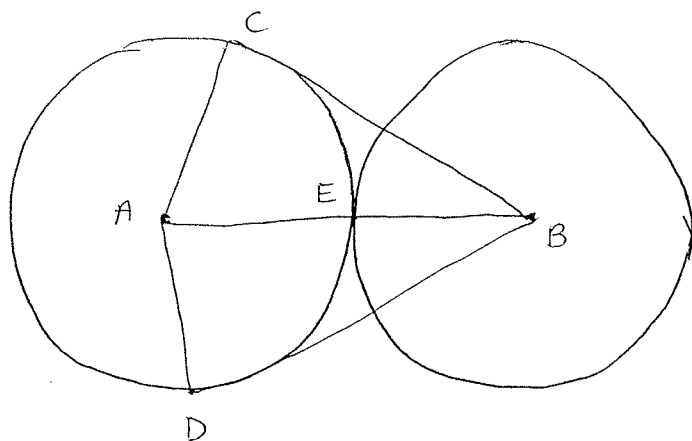
(ii) $\sqrt{2} \cos x - \sqrt{2} \sin x = 2 \cos(x + \frac{\pi}{4})$



(iii) $k = \sqrt{2}$

Endpoints 1 mark.
 Max/min 1 mark.

(c)



- ii) $\angle ACB = 90^\circ$ (radius is perpendicular to tangent at point of contact)
 Sim, $\angle ADB = 90^\circ$
 Since opposite angles $\angle ACB, \angle ADB$ are supplementary,
 then $ADBC$ is cyclic.

- (ii) AEB is a straight line - line joining the centres of 2 circles passes through their point of contact.

$$\angle ADB = 90^\circ$$

$\therefore AB$ is a diameter ($\angle$ in a semicircle is a rightangle)

$$\therefore AE = EB = DE = CE.$$

- (iii) $\triangle ADE$ is equilateral as all sides equal from (ii)

$$\therefore \angle DAE = 60^\circ$$

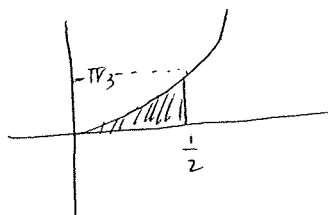
$$\therefore \angle DBA = 30^\circ \text{ (angle sum of } \triangle ABD = 180^\circ)$$

$$\text{Sim } \angle CBA = 30^\circ.$$

$$(d) (i) f\left(\frac{1}{2}\right) = 2\sin^{-1}\left(\frac{1}{2}\right)$$

$$= 2 \times \frac{\pi}{6}$$

$$= \frac{\pi}{3}$$



$$(ii) \frac{y}{2} = \sin^{-1}x$$

$$\therefore \sin \frac{y}{2} = x$$

$$A = \text{Area of rectangle} - \int_0^{\pi/3} x \, dy$$

$$= \left(\frac{\pi}{3} \times \frac{1}{2}\right) - \int_0^{\pi/3} \sin \frac{y}{2} \, dy$$

$$= \frac{\pi}{6} + 2 \left[\cos \frac{y}{2} \right]_0^{\pi/3}$$

$$\frac{\pi}{6} + 2 \left[\cos \frac{\pi}{6} - \cos 0 \right]$$

$$= \frac{\pi}{6} + 2 \left(\frac{\sqrt{3}}{2} - 1 \right)$$

$$= \left(\frac{\pi}{6} + \sqrt{3} - 2 \right) u^2$$

Area of rect = 1 mark

Integration = 1 mark

$$\begin{aligned}
 13. (a) (i) \quad m_{PQ} &= \frac{aq^2 - ap^2}{2aq - 2ap} \\
 &= \frac{a(q-p)(q+p)}{2a(q-p)} \\
 &= \frac{q+p}{2}
 \end{aligned}$$

$$\begin{aligned}
 \text{Eqn PQ : } y - ap^2 &= \frac{(p+q)}{2} (x - 2ap) \\
 2y - 2ap^2 &= (p+q)x - 2ap^2 - apq \\
 y &= \frac{(p+q)}{2}x - apq
 \end{aligned}$$

But PQ is a focal chord
 $\therefore (0, a)$ satisfies eqn.

$$\begin{aligned}
 a &= 0 - apq \\
 \therefore pq &= -1
 \end{aligned}$$

(ii) Eqn QR

$$\begin{aligned}
 y &= \frac{x^2}{4a} \\
 \frac{dy}{dx} &= \frac{x}{2a} \\
 \text{At } x &= 2aq \\
 m &= q
 \end{aligned}$$

$$\begin{aligned}
 \text{Eqn QR : } y - aq^2 &= q(x - 2aq) \\
 y - aq^2 &= qx - 2aq^2 \\
 y &= qx - aq^2 \quad \text{--- (1)}
 \end{aligned}$$

Coords R : Sub $x = 2ap$ into (1)

$$\begin{aligned}
 y &= 2apq - aq^2 \\
 \text{But } pq &= -1 \\
 y &= -2a - aq^2 \\
 \therefore R &\left(2ap, -2a - aq^2 \right) \\
 \text{i.e. } R &\left(-\frac{2a}{q}, -2a - aq^2 \right)
 \end{aligned}$$

Locus R : $x = -\frac{2a}{q}$

Sub $q = -\frac{2a}{x}$ into y

$$y = -2a - aq^2$$

$$y = -2a - a\left(-\frac{2a}{x}\right)^2$$

$$y = -2a - a\left(\frac{4a^2}{x^2}\right)$$

$$\boxed{y = -2a - \frac{4a^3}{x^2}}$$

(b) $\ddot{x} = 4x - 4$

(i) $\frac{d}{dx}\left(\frac{1}{2}v^2\right) = \ddot{x}$

$$\frac{d}{dx}\left(\frac{1}{2}v^2\right) = 4x - 4$$

$$\frac{1}{2}v^2 = \int (4x - 4) dx$$

$$\frac{1}{2}v^2 = 2x^2 - 4x + C$$

$$x = 6, v = 8$$

$$32 = 2 \times 36 - 4 \times 6 + C$$

$$32 = 72 - 24 + C$$

$$-16 = C$$

$$\therefore \frac{1}{2}v^2 = 2x^2 - 4x - 16$$

$$v^2 = 4x^2 - 8x - 32$$

(ii) $4x^2 - 8x - 32 \geq 0$

$$x^2 - 2x - 8 \geq 0$$

$$(x - 4)(x + 2) \geq 0$$



$x \leq -2, x \geq 4$ OR $x \geq 4$ only as initially $x = 6$.

(c) (i) LHS = $\frac{dT}{dt}$

$$= Ake^{-kt}$$

$$\text{RHS} = k(15 - T)$$

$$= k(15 - (15 - Ae^{-kt}))$$

$$= k(15 - 15 + Ae^{-kt})$$

$$= Ake^{-kt}$$

$$\text{LHS} = \text{RHS}.$$

$$(i) T = 15 - Ae^{-kt}$$

$$t=0, T = -5$$

$$-5 = 15 - Ae^0$$

$$\boxed{A = 20}$$

$$(ii) \frac{dT}{dt} = 5 \text{ when } t=0$$

$$\frac{dT}{dt} = Ake^{-kt}$$

$$5 = 20ke^0$$

$$k = \frac{1}{4}$$

$$(iii) 0 = 15 - 20e^{-0.25t}$$

$$\frac{-15}{-20} = e^{-0.25t}$$

$$\frac{3}{4} = e^{-0.25t}$$

$$\ln \frac{3}{4} = -0.25t$$

$$t = -4 \ln \frac{3}{4}$$

$$t = 1.15 \text{ min OR } 1 \text{ min } 9 \text{ s}$$

$$d) 9^{n+2} - 4^n \text{ is div. by } 5.$$

Step 1: Show true for $n=1$

$$9^3 - 4^1 = 725$$

725 is div. by 5

∴ True for $n=1$

Step 2: Assume true for $n=k$.

ie. $9^{k+2} - 4^k$ is div. by 5.

ie. $9^{k+2} - 4^k = 5M$ for M a pos. integer.

Step 3: Prove true for $n=k+1$

ie. $9^{k+3} - 4^{k+1}$ is div. by 5.

$$9 \cdot 9^{k+2} - 4 \cdot 4^k = 9(5M + 4^k) - 4 \cdot 4^k$$

$$= 45M + 9 \cdot 4^k - 4 \cdot 4^k$$

$$= 45M + 5 \cdot 4^k$$

$$= 5(9M + 4^k)$$



(From step 2

$$9^{k+2} = 5M + 4^k)$$

∴ This is integral for M, k pos. int.

14. (a) $x = t^2 e^{2-t}$

(i) $v = e^{2-t} \cdot 2t + t^2 \cdot (-1)e^{2-t}$
 $= 2te^{2-t} - t^2 e^{2-t}$

(ii) At rest initially then $t=0, v=0$

Sub $t=0$

$v = 0 - 0$

$v = 0$

(iii) At next $v=0$

$0 = 2te^{2-t} - t^2 e^{2-t}$

$0 = te^{2-t}(2-t)$

$\therefore t=0, t=2$

Next at rest when $t=2$

(iv) As $t \rightarrow \infty$

$x = t^2 e^{2-t}$
 $= \frac{t^2}{e^{t-2}}$

ie $x \rightarrow 0$

b) Let $f(x) = \log_e x - (x-1)^2$

$f'(x) = \frac{1}{x} - 2(x-1)$

$f(2) = \ln 2 - 1$

$f'(2) = \frac{1}{2} - 2 = -\frac{3}{2}$

$a = a_1 - \frac{f(a_1)}{f'(a_1)}$

$= 2 - \frac{(\ln 2 - 1)}{-\frac{3}{2}}$

$= 2 + 2 \frac{(\ln 2 - 1)}{3}$

$= 1.795$

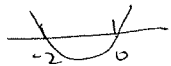
$= 1.80$

$$(c) \quad y = \log_e \left(\frac{2x}{2+x} \right)$$

$$(i) \quad \frac{2x}{2+x} > 0$$

$$\frac{2x(2+x)^2}{2+x} > 0 \quad (2+x)^2$$

$$2x(2+x) > 0$$



$$\boxed{x < -2, x > 0}$$

$$(ii) \quad 0 = \log_e \left(\frac{2x}{2+x} \right)$$

$$e^0 = \frac{2x}{2+x}$$

$$2+x = 2x$$

$$\boxed{2 = x}$$

$$(iii) \quad y = \log_e 2x - \log_e (2+x)$$

$$\frac{dy}{dx} = \frac{2}{2x} - \frac{1}{2+x}$$

$$= \frac{2+x-x}{x(2+x)}$$

$$= \frac{2}{x(2+x)}$$

For $x > 0$: $\frac{dy}{dx}$ is always positive

For $x < -2$: $\frac{2}{x(2+x)}$ is always positive.
 $\downarrow \quad \downarrow$ negative
 negative

$\therefore \frac{dy}{dx} > 0$ for $x > 0, x < -2$.

(iv) Inflexion Pts : $\frac{d^2y}{dx^2} = 0$ & change in concavity

$$\frac{d^2y}{dx^2} = \frac{x(2+x) \cdot 0 - 2 \cdot (2+2x)}{x^2(2+x)^2}$$

$$= \frac{-4(1+x)}{x^2(2+x)^2}$$

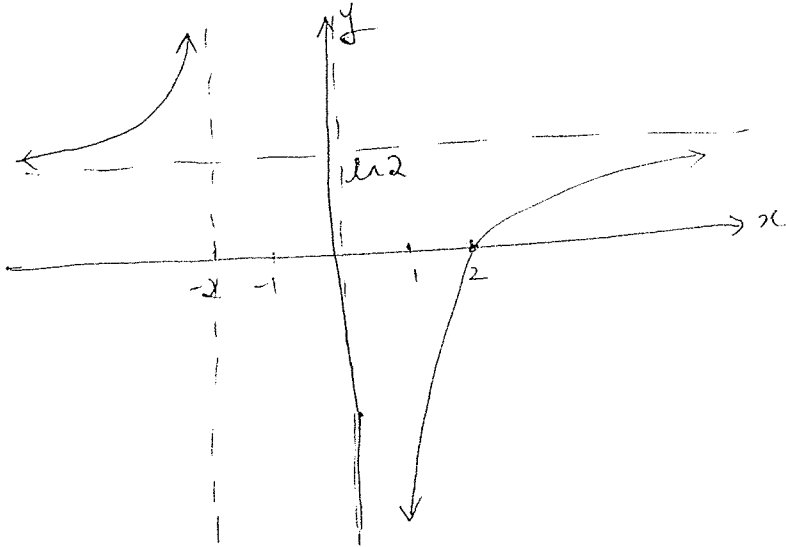
$$-4(1+x) = 0$$

$$\therefore x = -1.$$

But $x = -1$ is out of domain

$\therefore$ No inflexion points.

(v)



$$\text{As } x \rightarrow \infty : y = \ln 2x - \ln(2+x)$$

$$= \ln 2 + \ln x - \ln(2+x)$$

As $x \rightarrow \infty \Rightarrow \ln 2$ regardless of x
this cancels each other out.